SE444: Introduction to Artificial Intelligence
Makeup Mid-Term II Fall 2025

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Date: Wednesday, December 10, 2025 Time: 90 minutes

Instructions

- This is a closed-book exam. The University's code of ethics applies.
- Answer each exam question to the best of your knowledge.
- Clearly state any assumptions made that might be needed to understand your solution.
- Show your work – any sign of serious effort will be considered.
- No marks will be given for not making any effort.

Student Information

Student ID:	Student Name:

Summary of Exam Questions

Part	Topic	Time	Score	Mark	Mapped CLOs
Part I Multiple Choice	FOL — Chained Implications	5 min	1		CLO 1
	Propositional Logic — Satisfiability	5 min	1		CLO 1
	CSP — Arc Consistency	5 min	1		CLO 1, CLO 2
	Logical Agents — Environment Analysis	5 min	1		CLO 1, CLO 3
Part II Logical Reasoning	Translation to FOL	10 min	3		CLO 1
	Translation with Negation	10 min	3		CLO 1
	Negation and CNF Conversion	20 min	4		CLO 1
Part III CSP	Formulate Conference Scheduling as CSP	10 min	2		CLO 2
	Apply Arc Consistency (AC-3)	15 min	3		CLO 2
	Find a Consistent Solution	5 min	1		CLO 2
Total for Part I (Multiple Choice)		20 min	4		
Total for Part II (Logical Reasoning)		40 min	10		
Total for Part III (CSP)		30 min	6		
Total Points		90 min	20		

Part I. Critical-Thinking Multiple Choice Questions

Total Score: 4 points **Total Time:** 20 minutes

Instructions: Each question may have **one or more correct answers**. Select **all that apply**. Partial credit will be awarded for partially correct answers.

Answer Key Summary

Option	Q1 (5m)	Q2 (5m)	Q3 (5m)	Q4 (5m)
A	**X**		**X**	
B			**X**	**X**
C	**X**	**X**	**X**	
D				**X**
E			**X**	

Correct Answers:

- **Q1 (FOL):** A, C
- **Q2 (Propositional Logic):** C
- **Q3 (CSP):** A, B, C, E
- **Q4 (Agents):** B, D

Question I.1: First-Order Logic (FOL) — Chained Implications

Time: 5 minutes **Score:** 1 point **Mapped CLOs:** CLO 1

You are given the following FOL sentences (assume a non-empty domain):

1. $\forall x\,(R(x) \rightarrow S(x))$
2. $\forall x\,(S(x) \rightarrow T(x))$
3. $\exists x\,R(x)$

Which of the following sentences are logically entailed by (1), (2), and (3)?

A) $\exists x\,T(x)$

B) $\forall x\,T(x)$

C) $\exists x\,(R(x) \wedge T(x))$

D) $\forall x\,(T(x) \rightarrow R(x))$

E) $\exists x\,(S(x) \wedge \neg R(x))$

Answer: A and C

Explanation: From (3), $\exists x\, R(x)$. By (1), this x also satisfies $S(x)$. By (2), it also satisfies $T(x)$. So $\exists x\, T(x)$ (A) and $\exists x\, (R(x) \wedge T(x))$ (C) hold.

B is wrong because not all x need satisfy T. D is the converse of the chain. E is not entailed.

Question I.2: Propositional Logic — Satisfiability Analysis

Time: 5 minutes **Score:** 1 point **Mapped CLOs:** CLO 1

Consider a knowledge base (KB) with propositional atoms a, b, c, d, e:

1. $(a \wedge b) \rightarrow c$
2. $c \rightarrow d$
3. $d \rightarrow e$
4. $\neg e$
5. b

Which of the following statements is true?

A) KB is unsatisfiable.

B) KB $\models a$.

C) KB $\models \neg a$.

D) KB $\models c$.

E) KB can be satisfied with $a =$ true and $c =$ true.

Answer: C

Backward chain: (4) $\neg e \Rightarrow$ by (3), $\neg d \Rightarrow$ by (2), $\neg c \Rightarrow$ by (1) with (5) $b = T$, we need $\neg a$.

Model: $a = F, b = T, c = F, d = F, e = F$ satisfies all clauses.

KB is satisfiable (not A), forces $\neg a$ (C correct), forces $\neg c$ (not D), and $a = T, c = T$ contradicts (E incorrect).

Question I.3: Constraint Satisfaction Problems — Arc Consistency

Time: 5 minutes **Score:** 1 point **Mapped CLOs:** CLO 1, CLO 2

Consider a CSP with variables X, Y, Z.
Initial Domains: $D(X) = \{1, 2, 3\}$, $D(Y) = \{2\}$, $D(Z) = \{1, 2, 3\}$
Constraints: $X \neq Y$, $Y < Z$, $X < Z$
Which are true?

A) After AC, $D(Z)$ becomes $\{3\}$.

B) After AC, $D(X)$ becomes $\{1\}$.

C) After AC, all domains become singletons and the CSP is solved without search.

D) After AC, $D(X)$ still contains the value 2.

E) If we change "$Y < Z$" to "$Y \leq Z$", then $D(Z)$ would include $\{2, 3\}$ after AC.

Answer: A, B, C, E

AC-3 Process:

- $Y < Z$ with $Y = 2 \Rightarrow Z > 2 \Rightarrow D(Z) = \{3\}$ (A correct)
- $X \neq Y$ with $Y = 2 \Rightarrow X \neq 2$; $X < Z$ with $Z = 3 \Rightarrow X < 3 \Rightarrow D(X) = \{1\}$ (B correct)
- All singletons: $X = 1, Y = 2, Z = 3$ (C correct), D is wrong (2 removed)
- With $Y \leq Z$: $Z \geq 2 \Rightarrow D(Z) = \{2, 3\}$ (E correct)

Question I.4: Logical (Knowledge-Based) Agents — Environment Analysis

Time: 5 minutes **Score:** 1 point **Mapped CLOs:** CLO 1, CLO 3

A hospital deploys a **medical diagnosis robot** (partially observable, stochastic, sequential, dynamic).

A) Simple reflex agent sufficient if mapping is comprehensive...

B) Model-based reflex agent could track patient history...

C) Goal-based agent cannot operate rationally due to uncertainty...

D) Utility-based agent could weigh benefits/risks of diagnostic tests...

E) Learning agent provides no advantage since medical knowledge is fixed...

Answer: B, D

- A: Wrong. Simple reflex cannot handle partial observability.
- B: Correct. Model-based tracks internal state for better decisions.
- C: Wrong. Goal-based agents can handle uncertainty via probabilistic reasoning.
- D: Correct. Utility-based can value information and weigh risks.
- E: Wrong. Medical knowledge evolves; learning is beneficial.

Part II. Logical Reasoning in Healthcare Context

Total Score: 10 points **Total Time:** 40 minutes

Given Predicates: Doctor(x), Patient(x), Hospital(x), WorksAt(d,h), TreatedBy(p,d), HasDisease(p,dis), Disease(dis), Prescribes(d,p,m), Medication(m), Treats(m,dis), Examined(d,p)

Question II.1: Translation to FOL (3 pts)

"Every patient who is treated by a doctor at a hospital is prescribed at least one medication that treats one of their diseases."

Answer:

$\forall p, d, h\,(\text{Patient}(p) \wedge \text{Doctor}(d) \wedge \text{Hospital}(h) \wedge \text{TreatedBy}(p, d) \wedge \text{WorksAt}(d, h)) \rightarrow$
$(\exists m, dis\,(\text{Medication}(m) \wedge \text{Disease}(dis) \wedge \text{Prescribes}(d, p, m) \wedge \text{HasDisease}(p, dis) \wedge \text{Treats}(m, dis)))$

Question II.2: Translation with Negation (3 pts)

"No doctor prescribes medication to a patient they have not examined."

Answer:

$\forall d\,\forall p\,\forall m\,\big((\text{Doctor}(d) \wedge \text{Patient}(p) \wedge \text{Medication}(m) \wedge \text{Prescribes}(d, p, m)) \rightarrow \text{Examined}(d, p)\big)$

Or equivalently: prescription implies prior examination.

Question II.3: CNF Conversion (4 pts)

Negate and convert to CNF: $\forall x\,(\text{Doctor}(x) \rightarrow \text{Licensed}(x))$

Steps (1pt each):

1. Negate: $\neg\forall x\,(\text{Doctor}(x) \rightarrow \text{Licensed}(x))$
2. $\neg\forall \rightarrow \exists$: $\exists x\,\neg(\text{Doctor}(x) \rightarrow \text{Licensed}(x))$
3. Remove $\rightarrow$, De Morgan: $\exists x\,(\text{Doctor}(x) \wedge \neg\text{Licensed}(x))$
4. Skolemize: $\text{Doctor}(a) \wedge \neg\text{Licensed}(a)$

Part III. Constraint Satisfaction Problem (CSP)

Total Score: 6 points **Total Time:** 30 minutes

Context: Conference room scheduling. 3 rooms, 3 speakers.

Room	Time
Room 1	10:00–11:00 am
Room 2	10:30–11:30 am
Room 3	11:00–12:00 pm

Speaker	Available
Smith	Rooms 1, 2
Jones	Rooms 2, 3
Brown	Rooms 1, 2, 3

Question III.1: Formulate as CSP (2 pts)

Variables: R_1, R_2, R_3
Domains: $D(R_1) = \{\text{Smith}, \text{Brown}\}$, $D(R_2) = \{\text{Smith}, \text{Jones}, \text{Brown}\}$, $D(R_3) = \{\text{Jones}, \text{Brown}\}$
Constraints: $R_1 \neq R_2$ (overlap 10:30-11:00), $R_2 \neq R_3$ (overlap 11:00-11:30)

Question III.2: AC-3 Algorithm (3 pts)

Answer: NO PRUNING - All domains unchanged
All values have support under $\neq$ constraints since domains have 2+ values.
Final Domains: $D(R_1) = \{\text{Smith}, \text{Brown}\}$, $D(R_2) = \{\text{Smith}, \text{Jones}, \text{Brown}\}$, $D(R_3) = \{\text{Jones}, \text{Brown}\}$
CSP is arc-consistent but requires backtracking search.

Question III.3: Valid Solution (1 pt)

Valid Solutions (any one):

- $R_1 = \text{Smith}, R_2 = \text{Jones}, R_3 = \text{Brown}$
- $R_1 = \text{Smith}, R_2 = \text{Brown}, R_3 = \text{Jones}$
- $R_1 = \text{Brown}, R_2 = \text{Smith}, R_3 = \text{Jones}$
- $R_1 = \text{Brown}, R_2 = \text{Jones}, R_3 = \text{Brown}$